

SENTINEL CORE: REAL-WORLD IMPLEMENTATION STRATEGY

Project Technical Roadmap | Generated: 2026-01-28

This document outlines the transition path from the current research-based simulation to a scalable, cross-platform production environment.

1. The Application Layer (Cross-Platform Deployment)

To transform Sentinel into a downloadable application across all user devices, we utilize specialized software wrappers:

- **Desktop (Windows/Mac/Linux):** Integration via **Electron**. This converts the dashboard into a native executable while running the Python ML engine as a background 'Sidecar' process.
- **Mobile (iOS/Android):** Transformation into a **Progressive Web App (PWA)** or wrapping with **Capacitor**. This enables native features like Push Notifications for instant 'High-Risk' alerts.

2. Real-World Connectivity (The Data Switch)

Moving from simulation to the real world involves hot-swapping our 'Data Ingress' layer:

- **Direct Blockchain Ingress:** Replacing synthetic generators with live nodes (e.g., Alchemy, Infura, or Bitcoin Core). Our Kafka pipeline is designed to ingest live Mempool data without restructuring.
- **Architecture:** Hosting the ML Brain in **Docker Containers** on Kubernetes (AWS/Azure) to handle massive horizontal scaling during peak network traffic.

3. The SaaS Hierarchy

A tiered approach ensures both performance and accessibility:

- **The Edge:** Lightweight native apps for visualization.
- **The Intelligence:** FastAPI & ML Core performing complex manifold analysis in the cloud.
- **The Ledger:** Distributed Kafka clusters managing high-velocity data flow.

EXECUTIVE PITCH SUMMARY

'Sentinel is built on a Cloud-Native architecture. While today's demo showcases a digital twin simulation, the system is engineered specifically for scalability. By moving the Kafka clusters to the cloud and wrapping the dashboard in an Electron layer, we deliver a production-ready Security Firewall capable of protecting millions of transactions in real-time.'